

Dr. Doc:

Exam:-Biochemistry

Year:-2015 E.C

Dr. Doc

1.The conformational changes from the T (tense) to the R (relaxed) state are initiated by:

- A. Movement of the proximal histidine towards the heme
- B. Movement of the F-helix, which contains the proximal His
- C. Reorganization of protein-protein contacts between the heme
- D. Binding of oxygen to the heme

Answer D. Binding of oxygen to the heme

Hemoglobin exists in two conformations: the T (tense) state, which has a low affinity for oxygen, and the R (relaxed) state, which has a high affinity for oxygen. The transition from the T state to the R state is initiated when oxygen binds to the iron (Fe^{2+}) of the heme group. This binding triggers a series of structural changes that increase the oxygen affinity of the remaining subunits, producing cooperative binding.

Why Not Other Options?

A. Movement of the proximal histidine towards the heme ✗

Incorrect. This movement occurs after oxygen binds to the heme and is part of the structural changes triggered by oxygen binding.

B. Movement of the F-helix, which contains the proximal His ✗

Incorrect. The F-helix moves as a result of oxygen binding, not as the initiating event.

C. Reorganization of protein-protein contacts between the heme ✖

Incorrect. The reorganization occurs between the globin subunits, following oxygen binding, and helps stabilize the R state

2. In sickle cell anemia, the defect lies in which polypeptide?

A. Alpha chain

B. Beta chain

C. Gamma chain

D. Delta chain

Answer B. Beta chain

Sickle cell anemia is an autosomal recessive hemoglobinopathy caused by a point mutation in the β -globin (beta) chain of hemoglobin. In this mutation, glutamic acid is replaced by valine at the 6th position of the β -globin chain, producing hemoglobin S (HbS). Under low oxygen tension, HbS polymerizes, causing red blood cells to become sickle-shaped, leading to hemolysis, vaso-occlusion, and chronic anemia.

Why Not Other Options?

A. Alpha chain ✖

Incorrect. Defects in the α -globin chain are associated with α -thalassemia, not sickle cell anemia.

C. Gamma chain ✖

Incorrect. The γ -globin chain is a component of fetal hemoglobin (HbF) and is not affected in sickle cell anemia.

D. Delta chain ❌

Incorrect. The δ -globin chain forms part of hemoglobin A₂ (HbA₂) and is not involved in sickle cell disease.

3. In hemoglobin, the innate affinity of heme for carbon monoxide is diminished by the presence of:

A. His E7

B. His F9

C. Gly B6

D. Thr C4

Answer A. His E7

The distal histidine (His E7) in the globin chain reduces the naturally high affinity of heme for carbon monoxide (CO). It creates steric hindrance, preventing CO from binding in its preferred linear orientation. This favors oxygen binding and reduces CO affinity from about 25,000-fold to approximately 200–250-fold greater than oxygen. Although CO still binds more strongly than oxygen, the presence of His E7 significantly decreases its affinity.

Why Not Other Options?

B. His F9 ❌

Incorrect. The proximal histidine (His F8), not F9, binds directly to the iron (Fe^{2+}) of the heme and anchors the heme group. It does not primarily reduce CO affinity.

C. Gly B6 ❌

Incorrect. Glycine at position B6 is not involved in regulating oxygen or carbon monoxide binding to heme.

D. Thr C4 ✖

Incorrect. Threonine at position C4 does not play a significant role in heme-ligand binding.

4. The walls of the alveoli are composed of two types of cells, type I and type II. The function of type II is to:

A. Secrete surfactant

B. Trap dust and other debris

C. Replace mucus in the alveoli

D. Protect the lungs from bacterial invasion

Answer A. Secrete surfactant

Type II alveolar cells (type II pneumocytes) are cuboidal epithelial cells that secrete pulmonary surfactant, a phospholipid-rich substance that reduces surface tension within the alveoli. This prevents alveolar collapse (atelectasis), especially during expiration, improves lung compliance, and reduces the work of breathing. Type II pneumocytes also serve as progenitor cells, replacing damaged type I and type II alveolar cells after injury.

Why Not Other Options?

B. Trap dust and other debris ✖

Incorrect. Dust and debris are removed mainly by alveolar macrophages (dust cells), not type II pneumocytes.

C. Replace mucus in the alveoli ✖

Incorrect. Normal alveoli do not contain mucus. Mucus is produced by goblet cells and submucosal glands in the conducting airways.

D. Protect the lungs from bacterial invasion ❌

Incorrect. Although surfactant has some antimicrobial properties, the primary defense against bacteria in the alveoli is provided by alveolar macrophages

5. Carbon monoxide poisoning decreases O₂ delivery and causes tissue hypoxia and may lead to death in untreated individuals by:

A. Competitively binds to the O₂-binding site of hemoglobin at lower affinity

B. Reducing the ability of the hemoglobin molecule to release oxygen bound to other O₂-binding sites

C. Competitively interferes with the binding of 2,3-BPG to hemoglobin

D. All of the above

Answer B. Reducing the ability of the hemoglobin molecule to release oxygen bound to other O₂-binding sites

Carbon monoxide (CO) binds to the heme iron (Fe²⁺) of hemoglobin with an affinity about 200–250 times greater than oxygen. When CO binds to one or more heme groups, it increases the affinity of the remaining heme sites for oxygen, shifting the oxygen-hemoglobin dissociation curve to the left. As a result, hemoglobin holds onto oxygen more tightly, reducing oxygen release to the tissues and causing tissue hypoxia.

Why Not Other Options?

A. Competitively binds to the O₂-binding site of hemoglobin at lower affinity ❌

Incorrect. CO binds to the same heme iron as oxygen but with a much higher affinity, not a lower affinity.

C. Competitively interferes with the binding of 2,3-BPG to hemoglobin ❌

Incorrect. CO poisoning primarily affects oxygen binding and release. It does not act by competitively blocking the 2,3-BPG binding site.

D. All of the above ✖

Incorrect. Since options A and C are incorrect, "All of the above" cannot be the correct answer

Exam:-Biochemistry

Year:-2016 E.C

Dr. Doc

1.Which of the following would most readily bind oxygen?

- A. Hemoglobin (Hb) with one oxygen molecule bound
- B. Hemoglobin (Hb) with no oxygen bound
- C. Hemoglobin (Hb) with three oxygen molecules bound
- D. Hemoglobin (Hb) with four oxygen molecules bound

Answer C. Hemoglobin (Hb) with three oxygen molecules bound

Hemoglobin exhibits positive cooperativity, meaning that the binding of one oxygen molecule increases the affinity of the remaining heme groups for oxygen. As more oxygen molecules bind, hemoglobin changes from the T (tense) state, which has low oxygen affinity, to the R (relaxed) state, which has high oxygen affinity. Therefore, hemoglobin with three oxygen molecules already bound has the highest affinity for binding the fourth oxygen molecule. Hemoglobin with four oxygen molecules is already fully saturated and cannot bind another oxygen molecule.

Why Not Other Options?

A. Hb with one oxygen molecule bound ✖

Incorrect. Binding of the first oxygen increases hemoglobin's affinity for additional oxygen, but its affinity is lower than hemoglobin with three oxygen molecules already bound.

B. Hb with no oxygen bound ❌

Incorrect. Deoxygenated hemoglobin is in the T state, which has the lowest affinity for oxygen. Binding the first oxygen molecule is the most difficult step.

D. Hb with four oxygen molecules bound ❌

Incorrect. Hemoglobin with four oxygen molecules is fully saturated and has no remaining oxygen-binding sites.

Key exam point

💡 Hemoglobin shows positive cooperative binding.

💡 Binding of each oxygen molecule increases the affinity for the next oxygen molecule.

💡 T state = Low oxygen affinity; R state = High oxygen affinity.

💡 This cooperative behavior produces the sigmoid (S-shaped) oxygen dissociation curve.

Remember

✓ 3 O₂ bound → Highest affinity for the 4th O₂

✓ 0 O₂ bound = Lowest affinity (T state)

✓ 4 O₂ bound = Fully saturated; cannot bind more oxygen

2. All are true about surfactant except:

- A. Surfactant proteins A & D play an important role in the innate lung defense barrier against pathogenic organisms.
- B. Surfactant protein A both neutralizes and enhances the clearance of pathogens while modulating the inflammatory response.
- C. SP-B and SP-C are hydrophilic membrane proteins that are required for the proper biophysical function of the lung.
- D. The SP-B & SP-C proteins help spread the surfactant across the surface of the lung tissue, aiding in lowering surface tension.

Answer C. SP-B and SP-C are hydrophilic membrane proteins that are required for the proper biophysical function of the lung.

Pulmonary surfactant is produced by type II pneumocytes and is composed mainly of phospholipids and surfactant proteins (SP-A, SP-B, SP-C, and SP-D). SP-B and SP-C are hydrophobic proteins, not hydrophilic. They are essential for the proper spreading and stability of surfactant over the alveolar surface, thereby reducing surface tension and preventing alveolar collapse during expiration.

Why Not Other Options?

A. Surfactant proteins A & D play an important role in the innate lung defense barrier against pathogenic organisms. ✖

Incorrect. This statement is true. SP-A and SP-D are hydrophilic collectins that contribute to innate immunity by binding microorganisms, enhancing their clearance, and regulating inflammation.

B. Surfactant protein A both neutralizes and enhances the clearance of pathogens while modulating the inflammatory response. ✖

Incorrect. This statement is true. SP-A promotes the removal of bacteria, viruses, and fungi by immune cells while helping regulate the inflammatory response in the lungs.

D. The SP-B & SP-C proteins help spread the surfactant across the surface of the lung tissue, aiding in lowering surface tension. ❌

Incorrect. This statement is true. SP-B and SP-C facilitate the rapid spreading and stability of surfactant, allowing it to effectively reduce alveolar surface tension.

Key exam point

💡 Surfactant is produced by type II pneumocytes.

💡 SP-A and SP-D are hydrophilic and function in innate immune defense.

💡 SP-B and SP-C are hydrophobic and are essential for surfactant spreading and reduction of surface tension.

💡 Surfactant prevents alveolar collapse (atelectasis) and increases lung compliance.

Remember

✓ SP-A & SP-D = Defense (hydrophilic)

✓ SP-B & SP-C = Surface tension reduction (hydrophobic)

✓ Type II pneumocytes → Surfactant production

3. What would happen to red blood cells if the heme group were removed from hemoglobin?

- A. Red blood cells would not be able to bind oxygen.
- B. Antioxidant protein function can be reduced, such as peroxidase and catalase.
- C. Red blood cells would be able to reproduce.
- D. All are the correct answer.

Answer A. Red blood cells would not be able to bind oxygen.

Hemoglobin consists of four globin chains, each containing a heme group with a central ferrous iron (Fe^{2+}) atom. It is the heme group, not the globin protein, that binds oxygen reversibly. If the heme group were removed, hemoglobin would lose its oxygen-binding sites, making red blood cells unable to transport oxygen from the lungs to the tissues. As a result, the body's cells would receive insufficient oxygen, leading to tissue hypoxia and impaired cellular respiration. Although the globin chains would remain, they cannot bind oxygen without the heme group.

Why Not Other Options?

- B. Antioxidant protein function can be reduced, such as peroxidase and catalase. ✗

Incorrect. Heme is indeed a component of enzymes such as catalase and peroxidase, which protect cells from oxidative damage. However, the question specifically asks about removing the heme group from hemoglobin. The primary and immediate consequence is the loss of oxygen transport, not impairment of these enzymes.

- C. Red blood cells would be able to reproduce. ✗

Incorrect. Mature red blood cells lack a nucleus, mitochondria, and ribosomes, so they cannot divide, synthesize proteins, or reproduce. Removal of the heme group does not change this.

- D. All are the correct answer. ✗

Incorrect. "All of the above" is incorrect because option C is false. Therefore, only option A correctly answers the question.

Key exam point

- ✦ The heme group contains Fe^{2+} , which binds oxygen reversibly.
- ✦ Globin provides the protein structure, while heme is responsible for oxygen binding.
- ✦ Each hemoglobin molecule contains four heme groups and can carry four oxygen molecules.
- ✦ Loss of heme = Loss of oxygen transport = Tissue hypoxia.

Remember

- ✓ Heme (Fe^{2+}) = Oxygen binding
- ✓ Hemoglobin = Heme + Globin
- ✓ No heme → No oxygen transport

4. Commonest cause of respiratory distress in preterm infants is:

- A. Congenital heart disease
- B. Pneumonia
- C. Respiratory Distress Syndrome (RDS)
- D. Esophageal atresia

Answer C. Respiratory Distress Syndrome (RDS)

Respiratory Distress Syndrome (RDS), also known as Hyaline Membrane Disease, is the most common cause of respiratory distress in preterm infants. It is caused by surfactant deficiency due to the immaturity of type II pneumocytes. Without adequate surfactant, alveolar surface tension increases, causing alveolar collapse (atelectasis) during expiration. This leads to impaired gas exchange, hypoxemia, increased work of breathing, and respiratory distress shortly after birth. The risk of RDS increases as gestational age decreases, especially in infants born before 34 weeks of gestation.

Why Not Other Options?

A. Congenital heart disease ❌

Incorrect. Congenital heart disease can cause respiratory distress in newborns, but it is less common than RDS in preterm infants and is usually associated with structural abnormalities of the heart.

B. Pneumonia ❌

Incorrect. Neonatal pneumonia can cause respiratory distress due to bacterial or viral infection, but it is not the most common cause in preterm infants. Surfactant deficiency remains the leading cause.

D. Esophageal atresia ❌

Incorrect. Esophageal atresia is a congenital malformation that presents with excessive salivation, choking, coughing during feeding, and inability to pass a nasogastric tube. Although aspiration may lead to respiratory problems, it is not the commonest cause of respiratory distress in preterm infants.

Key exam point

💡 RDS (Hyaline Membrane Disease) is the commonest cause of respiratory distress in preterm infants.

💡 It is caused by surfactant deficiency due to immature type II pneumocytes.

✦ Clinical features include tachypnea, grunting, nasal flaring, chest retractions, and cyanosis.

✦ Treatment includes antenatal corticosteroids, exogenous surfactant replacement, and CPAP or mechanical ventilation when needed.

Remember

- ✓ Preterm infant + Respiratory distress = Think RDS
- ✓ RDS = Surfactant deficiency = Type II pneumocyte immaturity
- ✓ Surfactant reduces alveolar surface tension and prevents alveolar collapse

5. ___ is the form in which carbon dioxide is non-covalently bound to the globin chain of hemoglobin and transported in the blood.

- A. Oxyhemoglobin
- B. Deoxyhemoglobin
- C. Carbaminohemoglobin
- D. Methemoglobin

Answer C. Carbaminohemoglobin

Carbaminohemoglobin is formed when carbon dioxide (CO_2) binds to the amino groups of the globin chains of hemoglobin (not to the heme iron). This binding is reversible and accounts for about 20–30% of CO_2 transport in the blood. Most CO_2 (about 70%) is transported as bicarbonate (HCO_3^-), while a smaller amount is dissolved in plasma. Deoxygenated hemoglobin binds CO_2 more readily than oxygenated hemoglobin, a phenomenon known as the Haldane effect, which facilitates CO_2 transport from tissues to the lungs.

Note: Although the question says "non-covalently bound," carbaminohemoglobin is actually formed by a reversible covalent bond between CO_2 and the terminal amino groups of the globin chains. The intended answer is still Carbaminohemoglobin.

Why Not Other Options?

A. Oxyhemoglobin ❌

Incorrect. Oxyhemoglobin is formed when oxygen binds to the heme iron (Fe^{2+}) of hemoglobin. It is responsible for transporting oxygen, not carbon dioxide.

B. Deoxyhemoglobin ❌

Incorrect. Deoxyhemoglobin is hemoglobin without bound oxygen. Although it has a higher affinity for CO_2 than oxyhemoglobin, it is not the specific form produced after CO_2 binds.

D. Methemoglobin ❌

Incorrect. Methemoglobin contains ferric iron (Fe^{3+}) instead of ferrous iron (Fe^{2+}), making it unable to bind oxygen effectively. It is not involved in normal CO_2 transport.

Key exam point

💡 Carbaminohemoglobin transports about 20–30% of CO_2 in blood.

💡 Most CO_2 ($\approx 70\%$) is transported as bicarbonate (HCO_3^-).

💡 CO_2 binds to the globin chain, not the heme group.

💡 Deoxygenated hemoglobin binds CO₂ more readily (Haldane effect).

Remember

- ✓ CO₂ + Globin = Carbaminohemoglobin
- ✓ O₂ binds heme, CO₂ binds globin
- ✓ Most CO₂ is transported as bicarbonate (HCO₃⁻)

6. Which of the following is a genetic disorder of blood caused by a single nucleotide mutation in the β-globin gene?

- A. Diabetes mellitus
- B. Methemoglobinemia
- C. Sickle cell anemia
- D. Gall bladder disease

Answer C. Sickle cell anemia

Sickle cell anemia is an autosomal recessive genetic disorder caused by a single nucleotide substitution (point mutation) in the β-globin gene (HBB). Specifically, the sixth codon changes from GAG to GTG, resulting in the replacement of glutamic acid with valine at the sixth position of the β-globin chain. This produces hemoglobin S (HbS) instead of normal hemoglobin A (HbA).

When oxygen levels are low, HbS molecules polymerize, causing red blood cells to become rigid and sickle-shaped. These abnormal cells can block small blood vessels, leading to vaso-occlusive crises, chronic hemolytic anemia, tissue ischemia, and organ damage. The sickled cells also have a shortened lifespan (10–20 days) compared with normal red blood cells (about 120 days).

Why Not Other Options?

A. Diabetes mellitus ✖

Incorrect. Diabetes mellitus is a metabolic disorder characterized by impaired insulin secretion, insulin action, or both. It is not caused by a single mutation in the β -globin gene.

B. Methemoglobinemia ✖

Incorrect. Methemoglobinemia is a condition in which the iron in hemoglobin is oxidized from Fe^{2+} to Fe^{3+} , reducing its ability to carry oxygen. It may be acquired or inherited, but it is not caused by the β -globin gene mutation responsible for sickle cell anemia.

D. Gall bladder disease ✖

Incorrect. Gallbladder disease includes conditions such as gallstones and cholecystitis. It is not a genetic blood disorder caused by a mutation in the β -globin gene.

Key exam point

✦ Sickle cell anemia is caused by a point mutation in the β -globin (HBB) gene.

✦ Glutamic acid is replaced by valine at position 6 of the β -globin chain.

✦ It is inherited in an autosomal recessive pattern.

✦ Deoxygenated HbS polymerizes, causing red blood cells to sickle and leading to hemolysis and vaso-occlusion.

Remember

- ✓ β -globin gene mutation $\rightarrow$ HbS $\rightarrow$ Sickle cell anemia
- ✓ Glu $\rightarrow$ Val at position 6 of the β -globin chain
- ✓ Autosomal recessive + Point mutation = Sickle cell disease

7. The walls of the alveoli are composed of two types of cells, type I and type II. The function of type II is to:

- A. Protect the lungs from bacterial invasion
- B. Trap dust and other debris
- C. Replace mucus in the alveoli
- D. Secrete surfactant

Answer. D. Secrete surfactant

Type II pneumocytes (type II alveolar cells) are cuboidal epithelial cells that produce and secrete pulmonary surfactant, a phospholipid-rich substance that reduces surface tension within the alveoli. By lowering surface tension, surfactant prevents alveolar collapse (atelectasis) during expiration, increases lung compliance, and reduces the work of breathing. Type II pneumocytes also serve as progenitor cells, capable of proliferating and differentiating into type I pneumocytes after lung injury to help repair the alveolar epithelium.

Why Not Other Options?

A. Protect the lungs from bacterial invasion ❌

Incorrect. Protection against bacteria is mainly carried out by alveolar macrophages and immune components such as surfactant proteins A and D, not by the primary function of type II pneumocytes.

B. Trap dust and other debris ❌

Incorrect. Dust and debris are primarily removed by the mucociliary escalator in the conducting airways and by alveolar macrophages within the alveoli. Type II pneumocytes do not perform this function.

C. Replace mucus in the alveoli ❌

Incorrect. Alveoli normally do not contain mucus, as mucus would interfere with gas exchange. Mucus is produced by goblet cells and submucosal glands in the conducting airways, not by type II pneumocytes.

Key exam point

💡 Type I pneumocytes cover about 95% of the alveolar surface and are responsible for gas exchange.

💡 Type II pneumocytes produce surfactant and can regenerate both type I and type II alveolar cells after injury.

💡 Surfactant reduces alveolar surface tension, preventing alveolar collapse and improving lung compliance.

💡 Surfactant deficiency in preterm infants causes Respiratory Distress Syndrome (RDS).

Remember

- ✓ Type I pneumocytes = Gas exchange
- ✓ Type II pneumocytes = Surfactant production + Alveolar repair
- ✓ Alveolar macrophages = Remove bacteria, dust, and debris

8. Hemoglobin unlike myoglobin shows:

- A. Cooperative effect of combining with O₂
- B. Sigmoidal curve of oxygen association
- C. Cooperative index of 81
- D. Hyperbolic curve of oxygen association

Answer B. Sigmoidal curve of oxygen association

Hemoglobin has four heme groups and exhibits positive cooperative binding of oxygen. This cooperative binding gives hemoglobin its characteristic sigmoidal (S-shaped) oxygen dissociation curve. In contrast, myoglobin has only one heme group, does not exhibit cooperativity, and therefore has a hyperbolic oxygen dissociation curve.

Although option A is scientifically true, the best single answer is B, because the sigmoidal oxygen association curve is the distinctive feature that directly differentiates hemoglobin from myoglobin in MCQs.

Why Not Other Options?

A. Cooperative effect of combining with O₂ ✗

Incorrect as the best answer. This is the mechanism behind hemoglobin's oxygen binding, but the characteristic feature commonly asked in exams is the sigmoidal oxygen dissociation curve.

C. Cooperative index of 81 ✗

Incorrect. Hemoglobin's Hill coefficient (cooperative index) is approximately 2.8–3.0, not 81.

D. Hyperbolic curve of oxygen association ❌

Incorrect. A hyperbolic curve is characteristic of myoglobin, which has a single oxygen-binding site and does not show cooperativity.

Key exam point

💡 Hemoglobin = Cooperative O₂ binding = Sigmoidal oxygen dissociation curve.

💡 Myoglobin = Non-cooperative O₂ binding = Hyperbolic oxygen dissociation curve.

💡 Hemoglobin has four heme groups; myoglobin has one.

💡 Cooperativity allows efficient oxygen loading in the lungs and unloading in tissues.

Remember

✓ Hemoglobin → Sigmoidal curve

✓ Myoglobin → Hyperbolic curve

✓ Cooperativity is the reason hemoglobin has a sigmoidal curve.

9. Hemoglobin is a buffer because of:

- A. Lysine residue
- B. Histidine residue
- C. Iron molecule
- D. Cooperativity

Answer B. Histidine residue

Hemoglobin acts as an important buffer in the blood primarily because of its numerous histidine residues. The imidazole group of histidine has a pK_a close to physiological pH, allowing it to accept or donate hydrogen ions (H^+) as needed. This helps maintain blood pH by buffering excess acids or bases. Deoxygenated hemoglobin is an even better buffer than oxygenated hemoglobin because it has a greater capacity to bind H^+ , contributing to the Bohr effect and facilitating carbon dioxide transport.

Why Not Other Options?

A. Lysine residue ❌

Incorrect. Although lysine is an amino acid present in hemoglobin, its side chain is not the primary contributor to buffering at physiological pH because its pK_a is much higher than blood pH.

C. Iron molecule ❌

Incorrect. The iron (Fe^{2+}) in the heme group is responsible for binding oxygen, not for buffering hydrogen ions or regulating blood pH.

D. Cooperativity ❌

Incorrect. Cooperativity refers to the increased affinity of hemoglobin for oxygen as successive oxygen molecules bind. It is responsible for the sigmoidal oxygen dissociation curve, not the buffering action of hemoglobin.

Key exam point

✦ Histidine residues are responsible for the buffering action of hemoglobin.

✦ The imidazole group of histidine can bind or release H^+ near physiological pH.

✦ Deoxyhemoglobin is a better buffer than oxyhemoglobin.

✦ Hemoglobin plays an important role in acid-base balance and the Bohr effect.

Remember

✓ Hemoglobin buffer → Histidine residues

✓ Iron (Fe^{2+}) → Oxygen binding

✓ Cooperativity → Sigmoidal O_2 dissociation curve

10. Symptoms of low levels of hemoglobin are:

A. Feelings of rage

B. Trouble breathing and weakness

C. Trouble breathing and fever

D. Weakness and excess hair growth

Answer B. Trouble breathing and weakness

Low hemoglobin (anemia) reduces the blood's ability to carry oxygen to body tissues. As a result, tissues receive less oxygen, leading to symptoms such as weakness, fatigue, shortness of breath (especially on exertion), dizziness, palpitations, and pallor. The severity of symptoms depends on the degree and rate of development of anemia.

Why Not Other Options?

A. Feelings of rage ❌

Incorrect. Mood changes are not typical or characteristic symptoms of low hemoglobin. The common manifestations are related to reduced oxygen delivery, such as fatigue and dyspnea.

C. Trouble breathing and fever ❌

Incorrect. Shortness of breath is a common symptom of anemia, but fever is not. Fever usually suggests an infection or inflammatory condition, not low hemoglobin itself.

D. Weakness and excess hair growth ❌

Incorrect. Weakness is a common symptom of anemia, but excess hair growth (hirsutism) is unrelated to low hemoglobin. Hair changes, when present in iron deficiency, are more likely to involve hair loss, not increased hair growth.

Key exam point

💡 Low hemoglobin = Anemia = Reduced oxygen-carrying capacity of blood.

💡 Common symptoms include fatigue, weakness, shortness of breath, dizziness, pallor, and palpitations.

💡 Severe anemia may cause tachycardia, chest pain, or heart failure in susceptible individuals.

💡 Diagnosis is confirmed by measuring hemoglobin concentration and evaluating red blood cell indices.

Remember

✓ Low Hb → Weakness + Fatigue + Shortness of breath

✓ Anemia = Decreased oxygen delivery to tissues

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1.What is the role of hemoglobin in carbon dioxide transport?

- A. It carries carbon dioxide dissolved in plasma.
- B. It breaks down carbon dioxide into its constituent elements.
- C. It converts carbon dioxide into a soluble form for transportation.
- D. It binds to carbon dioxide and releases it in the lungs.

Answer D. It binds to carbon dioxide and releases it in the lungs.

Hemoglobin helps transport CO₂ mainly by binding CO₂ to the amino groups of its globin chains, forming carbaminohemoglobin. This binding is reversible: hemoglobin picks up CO₂ in the tissues and releases it in the lungs, where CO₂ is exhaled. Hemoglobin also helps transport H⁺, which supports the conversion of CO₂ to bicarbonate in red blood cells.

Why Not Other Options?

A. It carries carbon dioxide dissolved in plasma ❌

Incorrect. A small amount of CO_2 is transported dissolved in plasma, but this is not the primary role of hemoglobin.

B. It breaks down carbon dioxide into its constituent elements ❌

Incorrect. Hemoglobin does not break CO_2 into carbon and oxygen. CO_2 is instead converted into bicarbonate (HCO_3^-) by carbonic anhydrase in red blood cells.

C. It converts carbon dioxide into a soluble form for transportation ❌

Incorrect. Carbonic anhydrase, not hemoglobin, catalyzes the conversion of CO_2 and water into carbonic acid, which subsequently forms bicarbonate.

Key exam point

💡 Hemoglobin binds CO_2 to globin → carbaminohemoglobin.

💡 Most CO_2 is transported as bicarbonate (HCO_3^-).

💡 CO_2 binding is reversible, allowing hemoglobin to release CO_2 in the lungs.

Remember

✓ $\text{CO}_2 + \text{globin} \rightarrow \text{Carbaminohemoglobin}$

✓ Most $\text{CO}_2 \rightarrow \text{Bicarbonate}$

✓ Tissues $\rightarrow \text{Hb picks up } \text{CO}_2$; lungs $\rightarrow \text{Hb releases } \text{CO}_2$

2. In deoxyhemoglobin (Hb), the Fe^{2+} is 5-coordinated to:

- A. Four nitrogens of heme and the proximal His of Hb
- B. Four nitrogens of heme and an O_2 molecule
- C. Two nitrogens of heme and three His residues in Hb
- D. Two nitrogens of heme and three water molecules

Answer A. Four nitrogens of heme and the proximal His of Hb

In deoxyhemoglobin, the Fe^{2+} ion is in a 5-coordinate state. It is attached to the four nitrogen atoms of the porphyrin ring and to the proximal histidine (His F8) of the globin chain. The sixth coordination position is unoccupied, allowing oxygen to bind when hemoglobin becomes oxygenated.

When O_2 binds to this vacant sixth position, Fe^{2+} moves slightly into the plane of the heme ring. This structural change is transmitted through the proximal histidine to the globin chains and contributes to the transition from the T (tense, low-affinity) state to the R (relaxed, high-affinity) state. This is an important basis for cooperative oxygen binding by hemoglobin.

Why Not Other Options?

B. Four nitrogens of heme and an O_2 molecule ✖

Incorrect. In this case, the iron would be 6-coordinate, because O_2 occupies the sixth coordination site. This describes oxyhemoglobin, not deoxyhemoglobin.

C. Two nitrogens of heme and three His residues in Hb ✖

Incorrect. The Fe^{2+} is coordinated by four heme nitrogens, not two. Also, only the proximal histidine directly binds to the iron; the other histidine residues do not.

D. Two nitrogens of heme and three water molecules ✖

Incorrect. The heme iron is attached to four porphyrin nitrogen atoms. In normal deoxyhemoglobin, the sixth coordination site is essentially vacant, rather than being occupied by three water molecules.

Key exam point

✦ Deoxyhemoglobin Fe^{2+} = 5-coordinate $\rightarrow$ 4 heme N + proximal His (F8).

✦ The sixth coordination site is available for O_2 binding.

✦ O_2 binding converts Fe^{2+} to a 6-coordinate state and promotes $\text{T} \rightarrow \text{R}$ transition.

✦ This structural change contributes to positive cooperativity in hemoglobin.

Remember

✓ DeoxyHb $\rightarrow$ 5-coordinate

✓ OxyHb $\rightarrow$ 6-coordinate

✓ Proximal His (F8) $\rightarrow$ directly binds Fe^{2+}

✓ O_2 $\rightarrow$ binds at the sixth coordination site

3. Which of the following statement is NOT true?

A. Hemoglobin acts as an intracellular buffer

B. The proximal and distal histidine residues lie on the same side of the heme ring

C. Exposure of the heme ring to water results in oxidation

D. Protoporphyrin IX is a tetrapyrrole ring

Answer B. The proximal and distal histidine residues lie on the same side of the heme ring

Hemoglobin contains proximal histidine (His F8) and distal histidine (His E7), but they are located on opposite sides of the heme plane. The proximal histidine directly coordinates with Fe^{2+} , while the distal histidine does not directly bind the iron. Instead, it helps stabilize bound O_2 and reduces the likelihood of CO binding in a linear configuration.

Why Not Other Options?

A. Hemoglobin acts as an intracellular buffer ❌

Incorrect. This statement is true. Hemoglobin is an important intracellular buffer in red blood cells because its amino acid residues, especially histidine residues, can bind or release H^+ .

C. Exposure of heme ring to water results in oxidation ❌

Incorrect. This statement is generally true. The hydrophobic environment of the heme pocket helps protect the iron from oxidation. Exposure of heme to water/aqueous environments promotes oxidation of Fe^{2+} to Fe^{3+} , producing methemoglobin, which cannot effectively bind oxygen.

D. Protoporphyrin IX is a tetrapyrrole ring ❌

Incorrect. This statement is true. Protoporphyrin IX consists of four pyrrole rings linked together to form the porphyrin structure, with Fe^{2+} inserted into the center to form heme.

Key exam point

💡 Proximal His (F8) → directly binds Fe^{2+} .

✦ Distal His (E7) → does not directly bind Fe^{2+} ; helps stabilize O_2 .

✦ Proximal and distal histidines are positioned on opposite sides of the heme plane.

✦ Heme = Protoporphyrin IX + Fe^{2+} .

Remember

✓ Proximal His → Fe^{2+} binding

✓ Distal His → O_2 stabilization

✓ Opposite sides of heme → Proximal vs distal His

4. Which of the following describes the Bohr Effect?

A. Lowering the pH results in the release of O_2 from oxyhemoglobin.

B. Increasing the pressure of CO_2 results in the release of O_2 from oxyhemoglobin.

C. Increasing the pH increases the T-form of hemoglobin.

D. All of the above

E. None of the above

Answer A. Lowering the pH results in the release of O_2 from oxyhemoglobin.

The Bohr effect refers to the decrease in hemoglobin's affinity for O_2 when CO_2 increases or pH decreases. In metabolically active tissues, CO_2 production increases, leading to increased H^+ concentration and lower pH. This stabilizes the T (tense) state of hemoglobin and promotes O_2 unloading to the tissues.

Why Not Other Options?

B. Increasing the pressure of CO₂ results in the release of O₂ from oxyhemoglobin ✖

This is also physiologically true and is closely related to the Bohr effect. However, the classic definition of the Bohr effect emphasizes the effect of increased CO₂/decreased pH on O₂ affinity. If the question requires one best answer, A is the more direct formulation.

C. Increasing the pH increases the T-form of hemoglobin ✖

Incorrect. Increasing pH favors the R (relaxed) state, which has a higher affinity for O₂. The T state is favored by decreased pH.

D. All of the above ✖

Incorrect because C is false.

E. None of the above ✖

Incorrect because A correctly describes the Bohr effect.

Key exam point

💡 Bohr effect = ↓ pH or ↑ CO₂ → ↓ Hb affinity for O₂ → ↑ O₂ unloading.

💡 Low pH favors T state (low O₂ affinity).

💡 High pH favors R state (high O₂ affinity).

Remember

- ✓ ↓ pH → T state → O₂ release
- ✓ ↑ CO₂ → O₂ unloading
- ✓ ↑ pH → R state → O₂ binding

5. What contains 2 alpha chains and 2 beta chains that assume a quaternary higher-order conformation?

- A. Maternal hemoglobin
- B. Myoglobin
- C. Immunoglobulin
- D. Fetal hemoglobin

Answer A. Maternal hemoglobin

Maternal/adult hemoglobin (HbA) is composed of two α -globin chains and two β -globin chains ($\alpha_2\beta_2$). These four polypeptide chains interact to form the quaternary structure of hemoglobin. Each chain contains a heme group with Fe²⁺, so one HbA molecule can bind up to four O₂ molecules.

The arrangement of these four subunits is important for hemoglobin's cooperative oxygen binding. When one O₂ molecule binds, it increases the affinity of the other subunits for O₂, producing the characteristic sigmoidal oxygen dissociation curve. HbA is the predominant hemoglobin in healthy adults.

Why Not Other Options?

B. Myoglobin ✗

Myoglobin consists of one globin chain and one heme group, making it a monomer. It therefore has no quaternary structure and does not show cooperative oxygen binding.

C. Immunoglobulin ❌

Immunoglobulins are antibodies made from heavy and light chains. They do not contain α - and β -globin chains and have a completely different structure and function.

D. Fetal hemoglobin ❌

Fetal hemoglobin (HbF) has two α chains and two γ chains ($\alpha_2\gamma_2$). HbF has a higher affinity for oxygen than HbA, which allows oxygen to move efficiently from maternal blood to fetal blood through the placenta.

Key exam point

💡 Adult/maternal HbA = $\alpha_2\beta_2$ → quaternary structure.

💡 Fetal HbF = $\alpha_2\gamma_2$ → higher O_2 affinity.

💡 Myoglobin = single globin chain → monomeric.

💡 Four globin chains mean hemoglobin has quaternary structure and cooperative O_2 binding.

Remember

✓ HbA → $\alpha_2\beta_2$ → Adult

- ✓ HbF $\rightarrow \alpha_2\gamma_2 \rightarrow$ Fetus
- ✓ Myoglobin $\rightarrow$ 1 chain $\rightarrow$ Monomer
- ✓ Hemoglobin $\rightarrow$ 4 chains $\rightarrow$ Quaternary structure

6. When binding with oxygen, what type of binding curve does hemoglobin have?

- A. Sigmoidal
- B. Hyperbolic
- C. Linear
- D. Both A and B

Answer A. Sigmoidal

Hemoglobin has a sigmoidal (S-shaped) oxygen-binding curve because it has four heme groups and demonstrates positive cooperativity. When the first O_2 molecule binds to one heme group, it increases the affinity of the remaining heme groups for O_2 . This produces progressively easier oxygen binding and gives the curve its characteristic S-shape.

In contrast, myoglobin has only one heme group and therefore shows a hyperbolic curve because it does not exhibit cooperative binding.

Why Not Other Options?

B. Hyperbolic ✗

Incorrect. The hyperbolic curve is characteristic of myoglobin, not hemoglobin.

C. Linear ✗

Incorrect. Hemoglobin's oxygen-binding relationship is not linear because oxygen binding is cooperative.

D. Both A and B ❌

Incorrect. Hemoglobin shows a sigmoidal curve, whereas myoglobin shows a hyperbolic curve.

Key exam point

💡 Hemoglobin → Sigmoidal O₂-binding curve → Positive cooperativity.

💡 Myoglobin → Hyperbolic O₂-binding curve → No cooperativity.

Remember

✓ Hb = S-shaped (Sigmoidal)

✓ Mb = Hyperbolic

✓ Cooperativity → Sigmoidal curve

7. Which of the following statements concerning the binding of oxygen by hemoglobin is NOT correct?

A. The Bohr effect results in a lower affinity for oxygen at low pH values.

B. Carbon dioxide increases the oxygen affinity of hemoglobin by binding to the C-terminal groups of the polypeptide chains.

C. The oxygen affinity of hemoglobin increases as the percentage saturation increases.

D. The hemoglobin tetramer binds only one molecule of 2,3-BPG.

E. Deoxyhemoglobin has a high affinity for protons (H⁺).

Answer: B. Carbon dioxide increases the oxygen affinity of hemoglobin by binding to the C-terminal groups of the polypeptide chains.

The statement is not correct because CO₂ binding to the N-terminal amino groups of the globin chains forms carbaminohemoglobin and decreases hemoglobin's affinity for O₂. This favors the T (tense) state of hemoglobin and promotes oxygen unloading in metabolically active tissues.

CO₂ also contributes indirectly to the Bohr effect. In tissues, increased CO₂ leads to increased H⁺ concentration and decreased pH. Both increased CO₂ and increased H⁺ favor O₂ release from hemoglobin. In the lungs, where CO₂ is removed, hemoglobin's affinity for O₂ increases, allowing efficient oxygen loading.

Why Not Other Options?

A. The Bohr effect results in a lower affinity for oxygen at low pH values. ❌

This is true. A decrease in pH increases H⁺ binding to hemoglobin, stabilizing the T state and reducing O₂ affinity. Therefore, more oxygen is released to the tissues.

C. The oxygen affinity of hemoglobin increases as the percentage saturation increases. ❌

This is true because hemoglobin shows positive cooperativity. Binding of the first O₂ molecule causes a conformational change that makes subsequent O₂ molecules bind more easily. This produces the characteristic sigmoidal oxygen-binding curve.

D. The hemoglobin tetramer binds only one molecule of 2,3-BPG. ❌

This is true. One molecule of 2,3-BPG binds to one hemoglobin tetramer, specifically in the central cavity of deoxyhemoglobin. It stabilizes the T state and decreases O₂ affinity, facilitating oxygen delivery to tissues.

E. Deoxyhemoglobin has a high affinity for protons (H⁺). ❌

This is true. Deoxyhemoglobin can bind H^+ more readily than oxyhemoglobin, allowing hemoglobin to act as an important buffer and contributing to CO_2 transport.

Key exam point

✦ CO_2 binding to globin $\rightarrow$ Carbaminohemoglobin $\rightarrow$ $\downarrow$ O_2 affinity $\rightarrow$ $\uparrow$ O_2 unloading.

✦ $\downarrow$ pH / $\uparrow$ H^+ $\rightarrow$ T state $\rightarrow$ $\downarrow$ O_2 affinity.

✦ 2,3-BPG binds 1:1 with deoxyhemoglobin $\rightarrow$ stabilizes T state.

✦ Deoxyhemoglobin binds H^+ more readily than oxyhemoglobin.

Remember

- ✓ $CO_2 \rightarrow$ promotes O_2 release
- ✓ Low pH $\rightarrow$ promotes O_2 release
- ✓ 2,3-BPG $\rightarrow$ T state $\rightarrow$ $\downarrow$ O_2 affinity
- ✓ DeoxyHb $\rightarrow$ better H^+ buffer

8. When will hemoglobin have the highest affinity for oxygen?

- A. When carbon monoxide levels are high
- B. When pH is high

C. When 2,3-bisphosphoglycerate levels are high

D. None of the above

Answer: B. When pH is high

Hemoglobin's oxygen affinity increases when pH increases. A high pH means a lower concentration of H^+ , which favors the R (relaxed) state of hemoglobin. The R state has a higher affinity for O_2 , so hemoglobin binds oxygen more readily. This is the opposite of the Bohr effect, where decreased pH promotes oxygen unloading.

Why Not Other Options?

A. When carbon monoxide levels are high ❌

Incorrect. CO binds strongly to hemoglobin and occupies oxygen-binding sites. Although CO can increase the affinity of the remaining sites for O_2 , it reduces overall oxygen delivery to tissues and is not the normal physiological condition for maximum useful O_2 binding.

C. When 2,3-bisphosphoglycerate levels are high ❌

Incorrect. Increased 2,3-BPG decreases hemoglobin's O_2 affinity by stabilizing the T state, promoting oxygen release to tissues.

D. None of the above ❌

Incorrect because high pH (B) clearly increases hemoglobin's oxygen affinity.

Key exam point

💡 $\uparrow$ pH $\rightarrow$ $\uparrow$ O_2 affinity $\rightarrow$ R state favored.

✦ ↓ pH → ↓ O₂ affinity → T state favored (Bohr effect).

✦ ↑ 2,3-BPG → ↓ O₂ affinity → more O₂ unloading.

Remember

- ✓ High pH = High O₂ affinity
- ✓ Low pH = Low O₂ affinity
- ✓ High 2,3-BPG = Low O₂ affinity

9. What will happen at high elevations?

- A. 2,3-BPG concentration increases and hemoglobin's affinity for oxygen decreases
- B. 2,3-BPG concentration decreases and hemoglobin's affinity for oxygen decreases
- C. 2,3-BPG concentration decreases and hemoglobin's affinity for oxygen decreases
- D. 2,3-BPG concentration increases and hemoglobin's affinity for oxygen increases

Answer: A. 2,3-BPG concentration increases and hemoglobin's affinity for oxygen decreases

At high altitude, atmospheric oxygen pressure is lower, causing reduced oxygen availability. In response, red blood cells increase production of 2,3-BPG. Increased 2,3-BPG binds to deoxyhemoglobin and stabilizes its T (tense) state, thereby decreasing hemoglobin's affinity for oxygen.

This shifts the oxygen–hemoglobin dissociation curve to the right, allowing hemoglobin to release more oxygen to tissues despite the lower oxygen availability at high altitude. This is an important physiological adaptation to chronic hypoxia.

Why Not Other Options?

B. 2,3-BPG decreases and Hb affinity decreases ❌

Incorrect. At high altitude, 2,3-BPG increases, not decreases.

C. 2,3-BPG decreases and Hb affinity decreases ❌

Incorrect. Both parts are inconsistent with the normal adaptation. 2,3-BPG rises to facilitate oxygen unloading.

D. 2,3-BPG increases and Hb affinity increases ❌

Incorrect. Increased 2,3-BPG decreases, rather than increases, hemoglobin's oxygen affinity.

Key exam point

💡 High altitude → hypoxia → ↑ 2,3-BPG → ↓ Hb–O₂ affinity → ↑ O₂ unloading.

💡 Increased 2,3-BPG causes a rightward shift of the O₂ dissociation curve.

💡 This adaptation improves oxygen delivery to tissues.

Remember

✓ High altitude → ↑ 2,3-BPG

✓ ↑ 2,3-BPG → ↓ O₂ affinity

- ✓ $\downarrow$ O₂ affinity $\rightarrow$ $\uparrow$ O₂ unloading
- ✓ Right shift = Better tissue oxygen delivery

10. Identify the phospholipid that possesses surfactant activity and is synthesized shortly before parturition in full-term infants. Its deficiency in the lungs can cause respiratory distress syndrome (RDS).

- A. Dipalmitoylphosphatidylethanolamine
- B. Dipalmitoylphosphatidylcholine
- C. Dipalmitoylphosphatidylserine
- D. None of the above

Answer: B. Dipalmitoylphosphatidylcholine (DPPC)

Dipalmitoylphosphatidylcholine (DPPC), also called dipalmitoyl lecithin, is the major phospholipid responsible for the surface-tension-lowering activity of pulmonary surfactant. It is produced by type II alveolar pneumocytes and increases significantly during the late stages of fetal lung development, particularly near term.

DPPC spreads across the air–liquid interface of the alveoli, reducing surface tension and preventing alveolar collapse during expiration. In premature infants, insufficient surfactant—especially DPPC—can result in neonatal respiratory distress syndrome (RDS), also called hyaline membrane disease.

Why Not Other Options?

A. Dipalmitoylphosphatidylethanolamine ✗

Incorrect. Although phosphatidylethanolamine is a membrane phospholipid, it is not the principal phospholipid responsible for pulmonary surfactant activity.

C. Dipalmitoylphosphatidylserine ✗

Incorrect. Phosphatidylserine is an important component of cell membranes but is not the major surfactant phospholipid responsible for reducing alveolar surface tension.

D. None of the above ❌

Incorrect because DPPC (option B) is the major pulmonary surfactant phospholipid.

Key exam point

💡 DPPC = major phospholipid of pulmonary surfactant.

💡 It reduces alveolar surface tension and prevents alveolar collapse.

💡 Type II pneumocytes → surfactant production.

💡 Surfactant deficiency in premature infants → neonatal RDS.

Remember

✓ DPPC = Dipalmitoylphosphatidylcholine = Pulmonary surfactant

✓ Surfactant ↓ surface tension → prevents alveolar collapse

✓ Prematurity + surfactant deficiency → RDS